

Ansh Shah

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EDUCATION

University of Illinois Urbana-Champaign

Champaign, IL

B.S. Mathematics & Computer Science; GPA: 3.86; Full Tuition Scholarship '25-'26

Expected Grad: May 2028

Coursework: Data Structures and Algorithms, Computer Architecture, High Freq. Trading Tech., Graph Theory, Linear Algebra, Probability & Stats

Awards: USACO Silver Medalist, American Computer Science League National Finalist

EXPERIENCE

UIUC Financial Technology Lab

Jun. 2026 – Present

Trading Technology Research Assistant

Chicago, IL

- Developed the software pipeline for a GNSS timing receiver, automating u-blox configuration, 72-hour raw UBX capture, RINEX conversion, and Precise Point Positioning of 246K+ epochs across RTKLIB for beamforming
- Benchmarked 9 filtering configurations over dual-frequency GPS L1/L5, Galileo E1/E5a, and BeiDou B2a observations, achieving 1.2 cm position accuracy for trade timestamp synchronization in HFT infrastructure.

Future Leaders Initiative (Hack4Impact)

Jan. 2026 – May 2026

Software Engineer

Champaign, IL

- Developed an offline-first child-safety incident reporting platform for UNICEF, spanning a Flutter mobile app, React/TypeScript admin dashboard, and FastAPI backend on Supabase Postgres.
- Built a full-stack analytics dashboard: designed REST endpoints, keyword search, and filtering across 5 provinces and 210 chiefdoms, and rendered 5,000+ results in React with interactive charts, tables, and summary metrics.
- Engineered a client-side media metadata pipeline feeding offline sync queue, with Alembic migrations and a Hive-backed queue screen for reports submitted without connectivity, reducing report processing time by 22%.

National Organization for Business and Engineering

Sep. 2025 – Dec. 2025

Software Engineer

Urbana, IL

- Built a full-stack apparel marketplace platform using Next.js, MongoDB, and Vercel with server-side rendering and optimized API routes across 10+ pages, enabling a student-to-student commerce platform.
- Designed a component-based UI system with Figma prototyping and Tailwind CSS implementation, reducing duplicate code by 35% through standardized design patterns and reusable interface components.

PROJECTS

RF Modulation Classifier with Real-Time C++ Inference | *PyTorch, ONNX Runtime, C++* 2026

- Trained a 2.2M-parameter 1D CNN on RadioML 2018.01A (2.55M IQ recordings, 24 modulation classes), achieving 92% accuracy above +10 dB SNR while characterizing performance across the full -20 to +30 dB range.
- Exported the model to ONNX Runtime and engineered a high-performance C++ inference pipeline sustaining 1,290 inferences/sec with 775 μ s average latency, 1.7 ms p99 latency, and 1000/1000 prediction parity with PyTorch.
- Instrumented per-stage latency profiling and benchmarked multithreaded producer-consumer optimizations, demonstrating model execution accounted for 99.7% of inference time.

NBA Sports Betting Analytics Plugin | *Python, PostgreSQL, Streamlit, NBA API* 2025

- Programmed a probabilistic prediction system for NBA player performance using weighted moving averages with exponential decay, incorporating opponent defensive ratings, rest-day adjustments, and home/away factors across 49,000+ historical games stored in normalized PostgreSQL database with 8-table schema.
- Designed data pipeline integrating NBA API with rate-limited requests, implementing efficient queries on 600+ players and 13+ statistical categories for real-time performance analysis.

Rental Property Listing Scraper | *Python, Cr4wl, Pandas* 2025

- Developed a scalable web scraping pipeline that extracted and structured 50,000+ rental listings into JSON for automated database ingestion, allowing for a 30% reduction in manual data entry
- Reduced collection time by 40% through concurrent scraping, request optimization, and duplicate detection.

TECHNICAL SKILLS

Languages: Python, C++, Java, JavaScript, Dart, HTML/CSS, LaTeX, Mathematica

Frameworks/Libraries: React, FastAPI, Flutter, NEXT, Scikit-Learn, PyTorch, NumPy, Pandas, Matplotlib, ONNX

Developer Tools: Git/GitHub, PostgreSQL, Linux, RTKLIB, Vercel, Android Studio